TUTORIAL 11

Coordinates, Matrix Representations, and Change of Basis

11.1 Rank-nullity for linear maps

PROPOSITION 11.1

Let $T : V \to W$ be linear, with V finite-dimensional.
Then

$$\dim V = \dim \ker T + \dim \operatorname{Im} T.$$

EXERCISE 11.1

(a) $T : \mathbb{R}^3 \to \mathbb{R}^3$, $\ker T = \operatorname{span}\{(1,-1,2)^T\}$, $\operatorname{Im} T = \operatorname{span}\{(2,0,1)^T\}$;

(b) $T : M_2(\mathbb{R}) \to \mathbb{R}^2$, $\ker T = \operatorname{span}\left\{\begin{pmatrix} 2 & -1 \\ 0 & 3 \end{pmatrix}\right\}$.

Please decide whether a linear map exists in each case.

WORKED SOLUTION

Neither map exists. Part (a) contradicts rank-nullity. In part (b), the image would have to be a three-dimensional subspace of a two-dimensional space.

(a) $\dim \ker T + \dim \operatorname{Im} T = 1 + 1 = 2 \neq 3.$

(b) $\dim \operatorname{Im} T = \dim M_2(\mathbb{R}) - \dim \ker T = 4 - 1 = 3 > 2.$

MOTIVATION AND GUIDING IDEA

Motivation. A representing matrix is a computational tool that depends on the ordered bases of the domain and codomain. Changing coordinates leaves the map unchanged but can greatly simplify repeated calculations. Always identify the input and output coordinate systems to avoid reversing a change of basis.

NOTATION AND CONVENTIONS

- For ordered bases $B = (u_1, \dots, u_n)$ of U and $C = (v_1, \dots, v_m)$ of V, the notation $[T]_C^B$ denotes an $m \times n$ matrix taking B-coordinates as input and returning C-coordinates.
- $S \circ T$ applies T first and S second; I_V is the identity map on V.

Created by Chen Gadi

 © 2026 Chen Gadi

- $P_{B\leftarrow B'} = [I_V]_B^{B'}$ takes $[v]_{B'}$ to $[v]_B$. The arrow indicates the direction of coordinate conversion.
- If B is an ordered basis of $\mathbb{F}^n$ and E is the standard basis, $S_B = [\, b_1 \ \cdots \ b_n \,] = [I]_E^B$ is the basis-column matrix. Hence $[v]_E = S_B[v]_B$.
- $[T]_B$ abbreviates $[T]_B^B$ for an operator $T : V \to V$. Square matrices A, D are similar if $D = P^{-1}AP$ for an invertible P.

DEFINITION 11.1

Let U, V be finite-dimensional vector spaces over the same field, with ordered bases B, C, and let $T : U \to V$ be linear. Its representing matrix is defined by its columns:

$$[T]_C^B = \big[\, [T(u_1)]_C \ \cdots \ [T(u_n)]_C \,\big], \qquad [T(u)]_C = [T]_C^B[u]_B.$$

For another ordered basis B' of the domain, the columns of $P_{B\leftarrow B'}$ are the B-coordinate vectors of the vectors of B'. Its inverse is $P_{B'\leftarrow B}$. If $R : V \to W$ is linear and D is an ordered basis of W, then

$$[R \circ T]_D^B = [R]_D^C[T]_C^B.$$

If T is an isomorphism, $[T^{-1}]_B^C = ([T]_C^B)^{-1}$. For an operator with standard matrix A, the matrix in the ordered basis B is $S_B^{-1}AS_B$. Domain and codomain bases must match at every matrix product.

MOTIVATION AND GUIDING IDEA

Construct a map from prescribed vector images, check compatibility with linear dependence relations, compute a representing matrix and change bases. The emphasis is on the connection between an abstract map, coordinates and matrix calculations.

11.2 Composition of maps

RECALL AND TOOLS

- Let U, V, W be vector spaces over a field $\mathbb{F}$.
- Let $T : U \to V$ and $S : V \to W$ be linear maps.
- The map $S \circ T : U \to W$, defined for $u \in U$ by $(S \circ T)(u) = S(T(u))$, is their **composition**.
- $S \circ T$ is linear over $\mathbb{F}$.
- If both maps are surjective, their composition is surjective. If both are injective, their composition is injective. In particular, a composition of isomorphisms is an isomorphism.

Created by Chen Gadi

 © 2026 Chen Gadi

EXERCISE 11.2

$$T : \mathbb{R}^2 \to M_2(\mathbb{R}), \qquad T(a,b)^T = \begin{pmatrix} 2a & a-b \\ a+b & 3b \end{pmatrix},$$

$$S : M_2(\mathbb{R}) \to \mathbb{R}_3[X], \qquad S\begin{pmatrix} q & r \\ s & t \end{pmatrix} = qX^2 + (r-2s)X + t.$$

Please compute $S \circ T$.

WORKED SOLUTION

Apply T first and substitute its four matrix entries into the formula for S.

$$(S \circ T)(a,b)^T = (2a)X^2 + \big((a-b) - 2(a+b)\big)X + 3b = 2aX^2 - (a+3b)X + 3b.$$

NOTATION AND CONVENTIONS

Notation guide. For an ordered basis $B = (b_1, \ldots, b_n)$, the notation $[v]_B = (c_1, \ldots, c_n)^T$ means $v = \sum_{j=1}^{n} c_j b_j$. A^{-1} denotes the inverse, A^T the transpose, and $(A \mid I_n)$ an augmented matrix. $R_i \leftrightarrow R_j$ interchanges rows; $R_i \leftarrow \lambda R_i$ requires $\lambda \neq 0$; $R_i \leftarrow R_i + \lambda R_j$ requires $i \neq j$. rank A is the number of pivots in a row-echelon form. Distinguish a matrix unit E_{ij} from an entire elementary matrix E.

11.3 Coordinates

DEFINITION 11.2

Let V be a vector space over a field $\mathbb{F}$ with ordered basis $B = (v_1, \ldots, v_n)$.
Every $v \in V$ has a unique representation as a linear combination of the basis vectors. Thus there are unique $\alpha_1, \ldots, \alpha_n \in \mathbb{F}$ such that

$$v = \alpha_1 v_1 + \cdots + \alpha_n v_n$$

The coefficients $\alpha_1, \ldots, \alpha_n$ are the **coordinates of** v **relative to** B.

The column vector $\begin{pmatrix} \alpha_1 \\ \vdots \\ \alpha_n \end{pmatrix} \in \mathbb{F}^n$ is the **coordinate vector of** v **relative to** B, denoted $[v]_B$.

EXERCISE 11.3

$$B = ((1,1,0)^T, (0,1,1)^T, (1,0,1)^T), \qquad v = (5,3,6)^T.$$

Please find $[v]_B$ and please check it by reconstructing v.

WORKED SOLUTION

The basis-column matrix has determinant 2, so it is a basis. Equating the components of the

Created by Chen Gadi

 © 2026 Chen Gadi

EXERCISE 11.3 CONTINUED

linear combination gives the three equations.

$$a(1,1,0)^T + b(0,1,1)^T + c(1,0,1)^T = (a+c, a+b, b+c)^T.$$

$$a+c=5, \quad a+b=3, \quad b+c=6 \implies a=1, \quad b=2, \quad c=4.$$

$$[v]_B = (1,2,4)^T, \qquad (1,1,0)^T + 2(0,1,1)^T + 4(1,0,1)^T = (5,3,6)^T.$$

Conversely, knowing a vector's coordinates in a chosen basis allows us to reconstruct the vector.

With the notation above, $v = \alpha_1 v_1 + \cdots + \alpha_n v_n$ **if and only if** $[v]_B = \begin{pmatrix} \alpha_1 \\ \vdots \\ \alpha_n \end{pmatrix}$.

EXERCISE 11.4

$$B = ((2,1)^T, (-1,1)^T), \qquad [v]_B = (3,-2)^T.$$

Please find v in standard coordinates.

WORKED SOLUTION

The coordinate entries are the coefficients of the ordered basis vectors.

$$v = 3(2,1)^T - 2(-1,1)^T = (6,3)^T + (2,-2)^T = (8,1)^T.$$

PROPOSITION 11.2

Let V have dimension n over a field $\mathbb{F}$, and let B be an ordered basis of V.
The coordinate map $T_B : V \to \mathbb{F}^n$, defined by $T_B(v) = [v]_B$, is an isomorphism. In particular, $V \cong \mathbb{F}^n$.

RECALL AND TOOLS

Let V be a vector space over $\mathbb{F}$, let B be an ordered basis with n vectors, and let $u_1, \ldots, u_m$ be vectors in V. Then:

1. $\{u_1, \ldots, u_m\}$ is linearly independent (in V) **if and only if** $\{[u_1]_B, \ldots, [u_m]_B\}$ is linearly independent (in $\mathbb{F}^n$).

2. $\{u_1, \ldots, u_m\}$ spans V **if and only if** $\{[u_1]_B, \ldots, [u_m]_B\}$ spans $\mathbb{F}^n$.

3. $\{u_1, \ldots, u_m\}$ is a basis of V **if and only if** $\{[u_1]_B, \ldots, [u_m]_B\}$ is a basis of $\mathbb{F}^n$.

Created by Chen Gadi

 © 2026 Chen Gadi

EXERCISE 11.5

$$p_1 = 1 + 2X, \qquad p_2 = X + 3X^2, \qquad p_3 = 1 + X + 2X^2.$$

Please prove that these polynomials are independent in $\mathbb{R}_2[X]$ using coordinates in $(1, X, X^2)$.

WORKED SOLUTION

The coordinate columns are therefore independent. The coordinate map is an isomorphism, so the polynomials are independent as well.

$$[p_1] = (1, 2, 0)^T, \quad [p_2] = (0, 1, 3)^T, \quad [p_3] = (1, 1, 2)^T.$$

$$\det \begin{pmatrix} 1 & 0 & 1 \\ 2 & 1 & 1 \\ 0 & 3 & 2 \end{pmatrix} = (2 - 3) + (6 - 0) = 5 \neq 0.$$

EXERCISE 11.6

$$U = \operatorname{span}\{U_1, U_2, U_3\}, \qquad V = \operatorname{span}\{V_1, V_2\} \subseteq M_2(\mathbb{R}),$$

$$U_1 = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad U_2 = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad U_3 = \begin{pmatrix} 0 & 0 \\ 1 & 2 \end{pmatrix}, \quad V_1 = \begin{pmatrix} 1 & 0 \\ 1 & 2 \end{pmatrix}, \quad V_2 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

Please find bases and dimensions for U, V, and $U \cap V$.

WORKED SOLUTION

In each list, the entries uniquely determine the coefficients, proving independence. Membership in U requires a relation between the second-row entries. Substituting a general element of V determines the intersection.

$$U = \left\{ \begin{pmatrix} a & b \\ c & 2c \end{pmatrix} : a, b, c \in \mathbb{R} \right\}, \qquad V = \left\{ \begin{pmatrix} r & s \\ r + s & 2r \end{pmatrix} : r, s \in \mathbb{R} \right\}.$$

$$\begin{pmatrix} r & s \\ r + s & 2r \end{pmatrix} \in U \iff 2r = 2(r + s) \iff s = 0.$$

$$B_U = (U_1, U_2, U_3), \quad \dim U = 3, \qquad B_V = (V_1, V_2), \quad \dim V = 2,$$

$$U \cap V = \operatorname{span}\{V_1\}, \qquad \dim(U \cap V) = 1.$$

EXERCISE 11.7 FOUNDATION

Let $w_1 = (1, 2, 0)^T$, $w_2 = (0, 1, -1)^T$, $e_3 = (0, 0, 1)^T$. Please prove that $B = (w_1, w_2, e_3)$ is a basis of $\mathbb{R}^3$. Please find the inverse of its column matrix by row reduction and please compute $[b]_B$ for $b = (3, 8, 5)^T$.

WORKED SOLUTION

The equation $aw_1 + bw_2 + ce_3 = 0$ gives $a = 0$, $2a + b = 0$, and $-b + c = 0$, so $a = b = c = 0$.

Created By Chen Gadi

 © 2026 Chen Gadi

EXERCISE 11.7 FOUNDATION CONTINUED

Three independent vectors in $\mathbb{R}^3$ form a basis.

$$A = \begin{pmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 0 & -1 & 1 \end{pmatrix}.$$

$$\left(\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & 0 & 0 \\ 2 & 1 & 0 & 0 & 1 & 0 \\ 0 & -1 & 1 & 0 & 0 & 1 \end{array}\right) \xrightarrow{R_2 \leftarrow R_2 - 2R_1} \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & -2 & 1 & 0 \\ 0 & -1 & 1 & 0 & 0 & 1 \end{array}\right)$$

$$\xrightarrow{R_3 \leftarrow R_3 + R_2} \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & -2 & 1 & 0 \\ 0 & 0 & 1 & -2 & 1 & 1 \end{array}\right).$$

$$A^{-1} = \begin{pmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ -2 & 1 & 1 \end{pmatrix}, \qquad [b]_B = A^{-1} \begin{pmatrix} 3 \\ 8 \\ 5 \end{pmatrix} = \begin{pmatrix} 3 \\ -6+8 \\ -6+8+5 \end{pmatrix} = \begin{pmatrix} 3 \\ 2 \\ 7 \end{pmatrix}.$$

Indeed, $3w_1 + 2w_2 + 7e_3 = (3, 6+2, -2+7)^T = (3, 8, 5)^T$. The representation is unique because B is a basis.

EXERCISE 11.8

Please determine whether $B = (x^2 - x,\ x + 2,\ 1)$ is a basis of $\mathbb{R}_2[x]$ and of $\mathbb{R}_3[x]$. In the first case, please find the coordinates of $p(x) = ax^2 + bx + c$.

WORKED SOLUTION

Write p as a combination with coefficients α, β, γ and compare coefficients:

$$ax^2 + bx + c = \alpha(x^2 - x) + \beta(x + 2) + \gamma.$$

$$\alpha = a, \quad -\alpha + \beta = b, \quad 2\beta + \gamma = c.$$

$$[p]_B = (a,\ a + b,\ c - 2a - 2b)^T.$$

For each a, b, c there is a unique coordinate triple, so B is a basis of $\mathbb{R}_2[x]$. Every combination has degree at most 2 and cannot equal x^3, so this is not a basis of $\mathbb{R}_3[x]$.

EXERCISE 11.9

Please determine whether $B = (x^2 + 2x,\ x - 3,\ 2)$ is a basis of $\mathbb{R}_2[x]$ and of $\mathbb{R}_3[x]$. In the first case, please find the coordinates of $p(x) = ax^2 + bx + c$.

WORKED SOLUTION

Write p as a combination with coefficients α, β, γ and compare coefficients:

$$ax^2 + bx + c = \alpha(x^2 + 2x) + \beta(x - 3) + 2\gamma.$$

$$\alpha = a, \quad 2\alpha + \beta = b, \quad -3\beta + 2\gamma = c.$$

$$[p]_B = (a,\ b - 2a,\ (c + 3b - 6a)/2)^T.$$

Created by Chen Gadi

 © 2026 Chen Gadi

EXERCISE 11.9 CONTINUED

For each a, b, c there is a unique coordinate triple, so B is a basis of $\mathbb{R}_2[x]$. Every combination has degree at most 2 and cannot equal x^3, so this is not a basis of $\mathbb{R}_3[x]$.

11.4 Representing a linear map by a matrix

DEFINITION 11.3

Let U be an n-dimensional vector space over $\mathbb{F}$ with ordered basis $B = (u_1, \ldots, u_n)$.
Let V be an m-dimensional vector space over the same field with ordered basis $C = (v_1, \ldots, v_m)$.
For a linear map $T : U \to V$, define the matrix

$$[T]_C^B = \begin{pmatrix} | & | & | & | \\ [T(u_1)]_C & [T(u_2)]_C & \cdots & [T(u_n)]_C \\ | & | & | & | \end{pmatrix} \in \mathbb{F}^{m \times n}$$

$[T]_C^B$ is the **matrix of T relative to the ordered bases B and C.**

Construction

First compute $T(u_i)$, the images of the basis vectors of the domain.

Then express each image in coordinates relative to C, and use these coordinate columns in the given order.

PROPOSITION 11.3

Let U, V be vector spaces over $\mathbb{F}$, with ordered bases B of U and C of V. For a linear map $T : U \to V$ and every $u \in U$,

$$[T(u)]_C = [T]_C^B \cdot [u]_B$$

Compute this matrix once for the chosen ordered bases. It then converts the coordinate vector of any new input into the coordinate vector of its image relative to C.

EXERCISE 11.10

$$V = M_3(\mathbb{R}), \qquad T(A) = 2(A - A^T),$$
$$U = \{A : A = A^T\}, \qquad W = \{A : A = -A^T\}.$$

Please prove linearity, please find the kernel and image and their dimensions, and please give a representing matrix in a basis formed by a basis of U followed by a basis of W.

WORKED SOLUTION

Here E_{ij} has a single 1 in position (i, j) and zeros elsewhere. The free matrix entries give the displayed bases. The map vanishes on symmetric matrices and multiplies skew-symmetric matrices by 4, giving the diagonal representation.

$$T(\alpha A + \beta B) = 2\big(\alpha A + \beta B - \alpha A^T - \beta B^T\big) = \alpha T(A) + \beta T(B).$$

Created by Chen Gadi

 © 2026 Chen Gadi

EXERCISE 11.10 CONTINUED

$$T(A)=0 \iff A=A^T, \qquad \ker T=U.$$

$$T(A)^T=-T(A), \qquad K\in W \implies T(K/4)=K.$$

$$\operatorname{Im} T=W, \qquad B_U=(E_{11},E_{22},E_{33},E_{12}+E_{21},E_{13}+E_{31},E_{23}+E_{32}),$$

$$B_W=(E_{12}-E_{21},E_{13}-E_{31},E_{23}-E_{32}), \qquad \dim U=6, \quad \dim W=3, \quad 6+3=9.$$

$$A=\frac{A+A^T}{2}+\frac{A-A^T}{2}\in U+W, \quad U\cap W=\{0\}, \qquad [T]_{(B_U,B_W)}=\operatorname{diag}(0,0,0,0,0,0,4,4,4).$$

EXERCISE 11.11

$$[T]_E=A=\begin{pmatrix}0&2&-1\\0&3&1\\0&0&4\end{pmatrix}, \qquad V=\mathbb{R}^3.$$

Please find nonzero vectors v,u satisfying $T(v)=0$ and $T(u)=4u$. Then please find w satisfying $T(w)=w+u$.

WORKED SOLUTION

The first column is zero, so e_1 lies in the kernel. Solving the second system gives an eigenvector for 4. Scaling that eigenvector by $\frac{1}{3}$ yields the last equation.

$$v=(1,0,0)^T, \qquad Av=0.$$

$$(A-4I)u=0 \iff -4u_1+2u_2-u_3=0, \quad -u_2+u_3=0.$$

$$u=(1,4,4)^T, \qquad Au=(4,16,16)^T=4u.$$

$$w=\tfrac{1}{3}u=(\tfrac{1}{3},\tfrac{4}{3},\tfrac{4}{3})^T, \qquad T(w)=\tfrac{4}{3}u=w+u.$$

EXERCISE 11.12

Let $V=\mathbb{C}[x]$, the vector space of all complex polynomials.
Define the linear operator $N:V\to V$ by

$$(Np)(x):=-(7-x)\,p'(x).$$

For each $\theta\in\mathbb{R}$, define $U_\theta:V\to V$ by

$$U_\theta:=\exp(-i\theta N^2)=\sum_{k=0}^{\infty}\frac{(-i\theta)^k}{k!}N^{2k}.$$

Please use the following two ordered bases:

$$E=(1,x,x^2,x^3,\dots), \qquad B=(7,\ (7-x),\ (7-x)^2,\ (7-x)^3,\dots).$$

1. Please prove that N and U_θ are linear.
2. For $n\geq 1$, set $b_n(x)=(7-x)^n$, and set $b_0(x)=7$. Please compute $N(b_n)$ and $N^2(b_n)$.
3. Please compute $U_\theta(b_n)$ for all $n\geq 0$ and please deduce the form of $[U_\theta]_B$.

Created by Chen Gadi

 © 2026 Chen Gadi

EXERCISE 11.12 CONTINUED

4. Please find the matrix of U_θ restricted to $P_3 = \text{span}\{1, x, x^2, x^3\}$ in the ordered basis $E_3 = (1, x, x^2, x^3)$.

WORKED SOLUTION

1. Differentiation is linear, so

$$N(\alpha p + \beta r) = -(7-x)(\alpha p' + \beta r') = \alpha N(p) + \beta N(r).$$

To justify the series, restrict to P_d, the polynomials of degree at most d. This space is invariant under N, so on P_d the series is the exponential series of a finite matrix, converging for every θ. These definitions agree as d increases and therefore define a map on $\mathbb{C}[x]$. Linearity follows by taking limits of the partial sums in each coordinate.

2. For $n \geq 1$,

$$b_n'(x) = \frac{d}{dx}(7-x)^n = n(7-x)^{n-1} \cdot (-1) = -n(7-x)^{n-1}.$$

so

$$N(b_n) = -(7-x)\, b_n'(x) = -(7-x)\big(-n(7-x)^{n-1}\big) = n(7-x)^n = nb_n.$$

and hence

$$N^2(b_n) = N(nb_n) = nN(b_n) = n \cdot nb_n = n^2 b_n.$$

Also $N(b_0) = N(7) = 0$, so $N^2(b_0) = 0$.

3. Since $N^{2k}(b_n) = n^{2k} b_n$ for $n \geq 1$, the series gives

$$U_\theta(b_n) = \sum_{k=0}^{\infty} \frac{(-i\theta)^k}{k!}\, n^{2k}\, b_n = \exp(-i\theta n^2)\, b_n.$$

For $n = 0$, $U_\theta(b_0) = b_0$.

Thus the matrix in B is diagonal:

$$[U_\theta]_B = \text{diag}\big(1,\ e^{-i\theta},\ e^{-i4\theta},\ e^{-i9\theta}, \dots\big).$$

4. Write $q_1 = e^{-i\theta}$, $q_4 = e^{-i4\theta}$ and $q_9 = e^{-i9\theta}$. On P_3, use b_0, b_1, b_2, b_3. Their images are $b_0, q_1 b_1, q_4 b_2, q_9 b_3$, respectively.

Expand these four polynomials in E_3:

$$b_0 = 7, \qquad b_1 = 7 - x, \qquad b_2 = (7-x)^2 = x^2 - 14x + 49,$$

$$b_3 = (7-x)^3 = -x^3 + 21x^2 - 147x + 343.$$

Let C be the matrix with coordinate columns $[b_0]_{E_3}, [b_1]_{E_3}, [b_2]_{E_3}, [b_3]_{E_3}$:

$$C = \begin{pmatrix} 7 & 7 & 49 & 343 \\ 0 & -1 & -14 & -147 \\ 0 & 0 & 1 & 21 \\ 0 & 0 & 0 & -1 \end{pmatrix}.$$

Created by Chen Gadi

 © 2026 Chen Gadi

EXERCISE 11.12 CONTINUED

This matrix is invertible. The formula $[U_\theta]_{E_3} = C\,\mathrm{diag}(1, q_1, q_4, q_9)C^{-1}$ gives the following matrix:

$$[U_\theta]_{E_3} = \begin{pmatrix} 1 & 7(1-q_1) & 49(1-2q_1+q_4) & 343(1-3q_1+3q_4-q_9) \\ 0 & q_1 & 14(q_1-q_4) & 147(q_1-2q_4+q_9) \\ 0 & 0 & q_4 & 21(q_4-q_9) \\ 0 & 0 & 0 & q_9 \end{pmatrix}.$$

EXERCISE 11.13

$$T : \mathbb{R}^3 \to \mathbb{R}^2, \qquad T(x,y,z)^T = (3x+2y-z, 2x-y+4z)^T.$$

Please compute the standard-basis matrix using the images of the basis vectors.

WORKED SOLUTION

The basis-vector images are the matrix columns. The final multiplication verifies the formula for every input.

$$T(e_1) = (3,2)^T, \quad T(e_2) = (2,-1)^T, \quad T(e_3) = (-1,4)^T,$$

$$[T] = \begin{pmatrix} 3 & 2 & -1 \\ 2 & -1 & 4 \end{pmatrix}, \qquad [T]\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 3x+2y-z \\ 2x-y+4z \end{pmatrix}.$$

EXERCISE 11.14

$$T : \mathbb{R}_2[X] \to \mathbb{R}^2, \qquad T(p) = (p(0), p(-1))^T,$$

$$B = (1, 2+X, 1+X+2X^2), \qquad C = ((1,-1)^T, (0,1)^T).$$

Please compute $[T]_C^B$ and please verify $[T(p)]_C = [T]_C^B[p]_B$ for $p = a + bX + cX^2$.

WORKED SOLUTION

Compute each image and convert it to C-coordinates. To verify the identity, first equate coefficients in the B-expansion and then multiply by the matrix.

$$[(r,s)^T]_C = (r, r+s)^T,$$

$$T(1) = (1,1)^T, \quad T(2+X) = (2,1)^T, \quad T(1+X+2X^2) = (1,2)^T,$$

$$M = [T]_C^B = \begin{pmatrix} 1 & 2 & 1 \\ 2 & 3 & 3 \end{pmatrix}.$$

$$p = \alpha + \beta(2+X) + \gamma(1+X+2X^2) \implies \gamma = c/2, \quad \beta = b - c/2, \quad \alpha = a - 2b + c/2.$$

$$[p]_B = \begin{pmatrix} a - 2b + c/2 \\ b - c/2 \\ c/2 \end{pmatrix}, \qquad M[p]_B = \begin{pmatrix} a \\ 2a - b + c \end{pmatrix}.$$

$$T(p) = (a, a-b+c)^T, \qquad [T(p)]_C = (a, 2a-b+c)^T.$$

Created by Chen Gadi

 © 2026 Chen Gadi

11.4.1 Change-of-coordinate matrices

RECALL AND TOOLS

Definition: Let V be a finite-dimensional vector space over $\mathbb{F}$, with ordered bases B, B'. The **change-of-coordinate matrix** $[I_V]_B^{B'}$ from B' to B is the representing matrix $[T]_B^{B'}$ for the **identity map** $T : V \to V, T(v) = v$ for every $v \in V$.

- The fundamental identity is $[v]_B = [I_V]_B^{B'}[v]_{B'}$.
- The identity map is denoted I_V, with $I_V(v) = v$.
- Its columns are the coordinate vectors of the vectors of B' relative to the ordered basis B.

EXERCISE **11.15**

$$B = (1 + X, 2 + 3X), \qquad B' = (3 + 4X, 1 + 2X), \qquad p = a + bX.$$

Please compute the transition matrix $[I]_B^{B'}$ and please verify $[p]_B = [I]_B^{B'}[p]_{B'}$.

WORKED SOLUTION

The basis matrices contain polynomial coefficients in $(1, X)$. The transition first converts to standard coordinates and then to B. The coordinate formulas and final product explicitly verify the identity.

$$S_B = \begin{pmatrix} 1 & 2 \\ 1 & 3 \end{pmatrix}, \quad S_{B'} = \begin{pmatrix} 3 & 1 \\ 4 & 2 \end{pmatrix}, \quad S_B^{-1} = \begin{pmatrix} 3 & -2 \\ -1 & 1 \end{pmatrix},$$

$$[I]_B^{B'} = S_B^{-1} S_{B'} = \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}.$$

$$[p]_B = \begin{pmatrix} 3a - 2b \\ -a + b \end{pmatrix}, \qquad [p]_{B'} = \begin{pmatrix} a - b/2 \\ -2a + 3b/2 \end{pmatrix},$$

$$\begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} a - b/2 \\ -2a + 3b/2 \end{pmatrix} = \begin{pmatrix} 3a - 2b \\ -a + b \end{pmatrix}.$$

11.4.2 Matrices of compositions and inverse maps

PROPOSITION **11.4**

Let U, V, W be finite-dimensional vector spaces over $\mathbb{F}$, with ordered bases A, B, C, respectively.
Let $S : U \to V$ and $T : V \to W$ be linear. Then

$$[T \circ S]_C^A = [T]_C^B [S]_B^A$$

Created by Chen Gadi

 © 2026 Chen Gadi

Corollary

If $T : V \to W$ is an **isomorphism** and B, C are ordered bases of V, W, respectively, then

$$[T^{-1}]_B^C = \left([T]_C^B\right)^{-1}$$

11.5 Five descriptions of a linear map

Define $T : \mathbb{R}^2 \to \mathbb{R}^2$ so that, for every

$$(x, y) \in \mathbb{R}^2$$

we have

$$T(x, y) = (2x + 2y,\ 2x - 2y).$$

We now describe this same map in several ways.

Method 1: A formula for a general vector

This is a **general formula**: it specifies $T(v)$ for arbitrary $v = (x, y) \in \mathbb{R}^2$, rather than just particular vectors:

$$T(x, y) = (2x + 2y,\ 2x - 2y).$$

It defines the map directly on its entire domain.

Method 2: Images of a basis

Choose the standard basis:

$$e_1 = (1, 0), \qquad e_2 = (0, 1).$$

Then

$$T(e_1) = T(1, 0) = (2, 2), \qquad T(e_2) = T(0, 1) = (2, -2).$$

For $(x, y) = xe_1 + ye_2$, extend linearly:

$$T(x, y) = xT(e_1) + yT(e_2) = x(2, 2) + y(2, -2) = (2x + 2y,\ 2x - 2y).$$

Thus specifying the images of a basis defines the same map.

Method 3: A matrix

Write the input as a column:

$$\begin{pmatrix} x \\ y \end{pmatrix}.$$

Define

$$A = \begin{pmatrix} 2 & 2 \\ 2 & -2 \end{pmatrix}.$$

Then

$$T(x, y) = A \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 2x + 2y \\ 2x - 2y \end{pmatrix}.$$

This matrix represents the same map in the standard basis.

Created by Chen Gadi

 © 2026 Chen Gadi

Method 4: Familiar linear operations

View T as performing the following operations:

- First component: twice the sum of the coordinates.
- Second component: twice the difference of the coordinates.

Thus

$$T(x, y) = \big(2\text{sum}(x, y), 2\text{diff}(x, y)\big),$$

These operations are linear, so T is linear.

Method 5: A structural decomposition

Specify that T satisfies

$$T(e_1 + e_2) = (4, 0), \qquad T(e_1 - e_2) = (0, 4),$$

and that T is linear.

Since $\{e_1+e_2, e_1-e_2\}$ is a basis of $\mathbb{R}^2$, these requirements determine T uniquely. Calculation gives the same formula:

$$T(x, y) = (2x + 2y,\ 2x - 2y).$$

Remarks

- A **general formula** defines $T(v)$ for an arbitrary vector v.
- A linear map may be specified by a formula, basis images, a matrix or a structural description.
- The descriptions differ, but the map itself is the same.

11.5.1 Summary of the methods

RECALL AND TOOLS

1. **An explicit formula for a general vector**
   Specify $T(v)$ for every $v \in V$.
   Check: the formula is well defined and satisfies both linearity conditions.

2. **Basis images and linear extension**
   If $B = \{v_1, \dots, v_n\}$ is a basis of V, freely choose

   $$T(v_1), \dots, T(v_n) \in W$$

   and define, for $v = \sum_{i=1}^{n} \alpha_i v_i$,

   $$T(v) := \sum_{i=1}^{n} \alpha_i\, T(v_i).$$

   Each choice of basis images determines exactly one linear map.

3. **A matrix**
   For $V = \mathbb{F}^n$ and $W = \mathbb{F}^m$, define

   $$T(x) = Ax$$

Created by Chen Gadi

 © 2026 Chen Gadi

where A is an $m \times n$ matrix.
Every matrix defines a linear map. After choosing ordered bases, every linear map between finite-dimensional spaces has a matrix representation.
Remark: the matrix depends on the chosen bases; the map itself does not.

4. **A familiar linear operation**
Use an operation such as differentiation, integration or multiplication by a fixed scalar.
Examples:
$$T(P) = P', \qquad T(f) = \int_{-1}^{2} f(x)\,dx.$$
Specify a domain and codomain, check well-definedness and verify linearity in each case.

5. **A direct-sum decomposition**
A map can be specified by its action on the components of a direct sum.
For example, if $V = U \oplus W$, define T by
$$T(u) = 0 \ \ (\forall u \in U), \qquad T(w) = w \ \ (\forall w \in W).$$
Linearity follows from the direct-sum decomposition and the linear action on each component.

Common errors

- Adding a nonzero constant vector destroys linearity: $T(v) = v + b$ with $b \neq 0$ is not linear.
- Always specify the domain and codomain: $T : V \to W$. The image $\operatorname{Im} T$ may be a proper subspace of the codomain.
- Distinguish a vector, its coordinate vector in an ordered basis, and the matrix representing a map.

Created by Chen Gadi

Additional practice

EXERCISE 11.16 FOUNDATION

$$T(x, y) = (2x - y, x + y), \quad B = ((1, 0)^T, (1, 1)^T), \quad C = ((1, 1)^T, (0, 1)^T), \quad v = (3, 2)^T.$$

Please compute $[T]_C^B$ and $[T^{-1}]_B^C$. Please verify the coordinate identity and please recover v using the inverse.

WORKED SOLUTION

Convert the images of the B-vectors to C-coordinates and place them in the columns. The inverse takes C-coordinates and returns B-coordinates. Reconstruction in B recovers $(3, 2)^T$.

$$[(r, s)^T]_C = (r, s - r)^T, \quad T(1, 0)^T = (2, 1)^T, \quad T(1, 1)^T = (1, 2)^T.$$

$$M = [T]_C^B = \begin{pmatrix} 2 & 1 \\ -1 & 1 \end{pmatrix}, \quad \det M = 3, \quad [T^{-1}]_B^C = M^{-1} = \frac{1}{3}\begin{pmatrix} 1 & -1 \\ 1 & 2 \end{pmatrix}.$$

 © 2026 Chen Gadi

EXERCISE 11.16 FOUNDATION CONTINUED

$$v = (1,0)^T + 2(1,1)^T, \quad [v]_B = (1,2)^T, \quad T(v) = (4,5)^T,$$

$$[T(v)]_C = (4,1)^T = M(1,2)^T, \qquad M^{-1}(4,1)^T = \frac{1}{3}(3,6)^T = (1,2)^T.$$

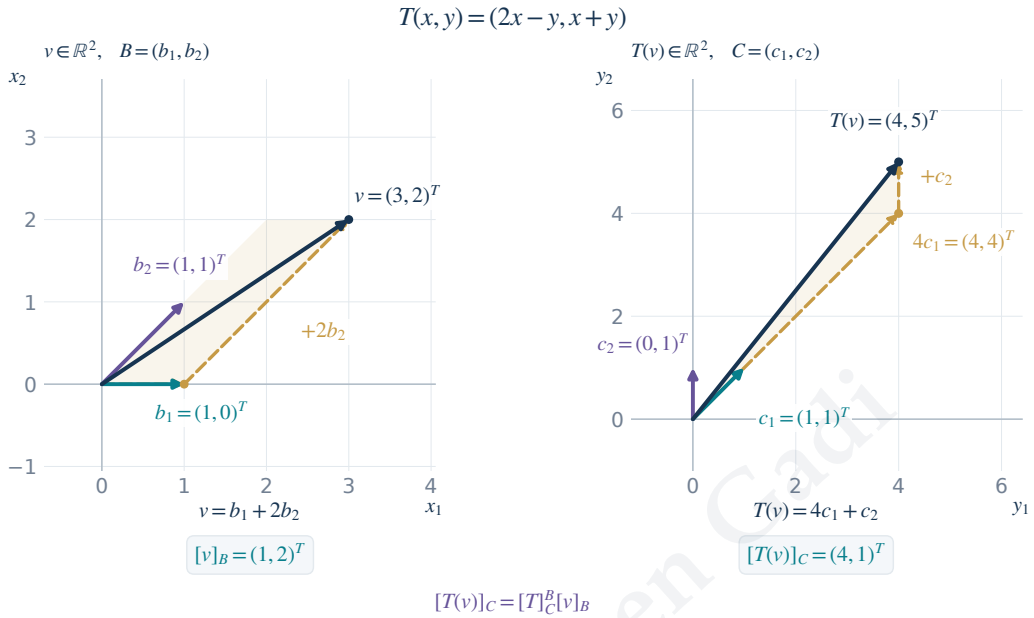


Figure 11.1. The arrows show vectors in standard coordinates; the gold paths show the decompositions $v = b_1 + 2b_2$ and $T(v) = 4c_1 + c_2$. Coordinate columns appear separately: $[v]_B = (1,2)^T$ and $[T(v)]_C = (4,1)^T$. The vector $T(v)$ itself is $(4,5)^T$.

11.6 Determinants and Change of Basis

RECALL AND TOOLS

We also use determinants over a field $\mathbb{F}$, such as $\mathbb{R}$, $\mathbb{C}$, or $\mathbb{Z}_p$ with p prime. Arithmetic in the formulas is performed in that field. In particular,

$$\det \begin{pmatrix} a & b \\ c & d \end{pmatrix} = ad - bc.$$

If $\mathcal{B}, \mathcal{C}$ are bases of the same space and the columns of S are the coordinates of the vectors of $\mathcal{C}$ in $\mathcal{B}$, then

$$[T]_{\mathcal{C}} = S^{-1}[T]_{\mathcal{B}} S.$$

Multiplicativity gives

$$\det[T]_{\mathcal{C}} = \det(S^{-1}) \det[T]_{\mathcal{B}} \det S = \det[T]_{\mathcal{B}}.$$

Here $T : V \to V$ is linear and $\dim V < \infty$. Thus we may define $\det T = \det[T]_{\mathcal{B}}$; this scalar does not depend on the choice of basis.

Created by Chen Gadi

 © 2026 Chen Gadi

EXERCISE 11.17

Let V be a real vector space with basis $\mathcal{B} = (b_1, b_2)$, and let $T : V \to V$ satisfy

$$T(b_1) = 4b_1 + b_2, \qquad T(b_2) = b_1 + 2b_2.$$

For $\mathcal{C} = (b_1, b_1 + b_2)$, please compute $[T]_{\mathcal{B}}$ and $[T]_{\mathcal{C}}$, and please verify that their determinants agree.

WORKED SOLUTION

The images of the basis vectors form the columns, so

$$A = [T]_{\mathcal{B}} = \begin{pmatrix} 4 & 1 \\ 1 & 2 \end{pmatrix}, \qquad \det A = 4 \cdot 2 - 1 \cdot 1 = 7.$$

Write $c_1 = b_1$, $c_2 = b_1 + b_2$. Then $b_2 = c_2 - c_1$, and hence

$$T(c_1) = 4c_1 + (c_2 - c_1) = 3c_1 + c_2,$$

$$T(c_2) = T(b_1) + T(b_2) = 5b_1 + 3b_2 = 2c_1 + 3c_2.$$

Therefore

$$[T]_{\mathcal{C}} = \begin{pmatrix} 3 & 2 \\ 1 & 3 \end{pmatrix}, \qquad \det[T]_{\mathcal{C}} = 3 \cdot 3 - 2 \cdot 1 = 7.$$

The change-of-basis calculation gives the same result:

$$S = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \qquad S^{-1}AS = \begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 4 & 5 \\ 1 & 3 \end{pmatrix} = \begin{pmatrix} 3 & 2 \\ 1 & 3 \end{pmatrix}.$$

EXERCISE 11.18

$$p(x) = a_0 + a_1 x + a_2 x^2 + a_3 x^3 \in \mathbb{F}_{\le 3}[x], \qquad \ell_1(p) = p(0), \quad \ell_2(p) = p(-1), \quad \ell_3(p) = p'(0).$$

Please find the matrix of the three maps in the basis $(1, x, x^2, x^3)$, row-space and column-space bases, and the rank. Please find all p satisfying $p(0) = 2$, $p(-1) = 3$, $p'(0) = 1$, and the dimension of the solution set.

WORKED SOLUTION

Expanding the evaluations and derivative gives the matrix. The displayed minor is invertible over every field, so the rank is 3 and the listed rows and columns are bases. The solution set is an affine line of dimension 1. Integer constants denote the corresponding field elements, so the calculation also holds in positive characteristic.

$$A = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 1 & -1 & 1 & -1 \\ 0 & 1 & 0 & 0 \end{pmatrix}, \qquad \det \begin{pmatrix} 1 & 0 & 0 \\ 1 & -1 & 1 \\ 0 & 1 & 0 \end{pmatrix} = -1.$$

$$B_{\text{row}} = \{(1,0,0,0), (1,-1,1,-1), (0,1,0,0)\}, \qquad B_{\text{col}} = \{(1,1,0)^T, (0,-1,1)^T, (0,1,0)^T\}.$$

$$a_0 = 2, \quad a_1 = 1, \quad 2 - 1 + a_2 - a_3 = 3 \implies a_2 = 2 + a_3.$$

Created by Chen Gadi

 © 2026 Chen Gadi

EXERCISE 11.18 CONTINUED

$$p(x) = 2 + x + 2x^2 + t(x^2 + x^3), \qquad t \in \mathbb{F}.$$

EXERCISE 11.19 FOUNDATION

$$T : \mathbb{R}_{\leq 2}[X] \to \mathbb{R}^2, \qquad T(p) = (p(2), p(-2))^T.$$

Please prove linearity, please find the kernel and image, and please check rank-nullity. Please construct a linear map S in the opposite direction with $T \circ S = I$. Please decide whether $S \circ T = I$.

WORKED SOLUTION

Evaluation preserves addition and scalar multiplication. The formula for S is linear and takes the required values at the two points, so T is onto. The reverse composition preserves the evaluations but loses the quadratic coefficient, so it is not the identity.

$$p = a + bX + cX^2 \in \ker T \iff a + 2b + 4c = 0, \quad a - 2b + 4c = 0 \iff b = 0,\ a = -4c.$$

$$\ker T = \operatorname{span}\{X^2 - 4\}, \quad \dim \ker T = 1, \qquad S((u,v)^T) = \frac{u+v}{2} + \frac{u-v}{4}X.$$

$$(T \circ S)(u,v)^T = (u,v)^T, \qquad \operatorname{Im} T = \mathbb{R}^2, \quad 1 + 2 = 3.$$

$$(S \circ T)(a + bX + cX^2) = a + 4c + bX, \qquad (S \circ T)(X^2) = 4 \neq X^2.$$

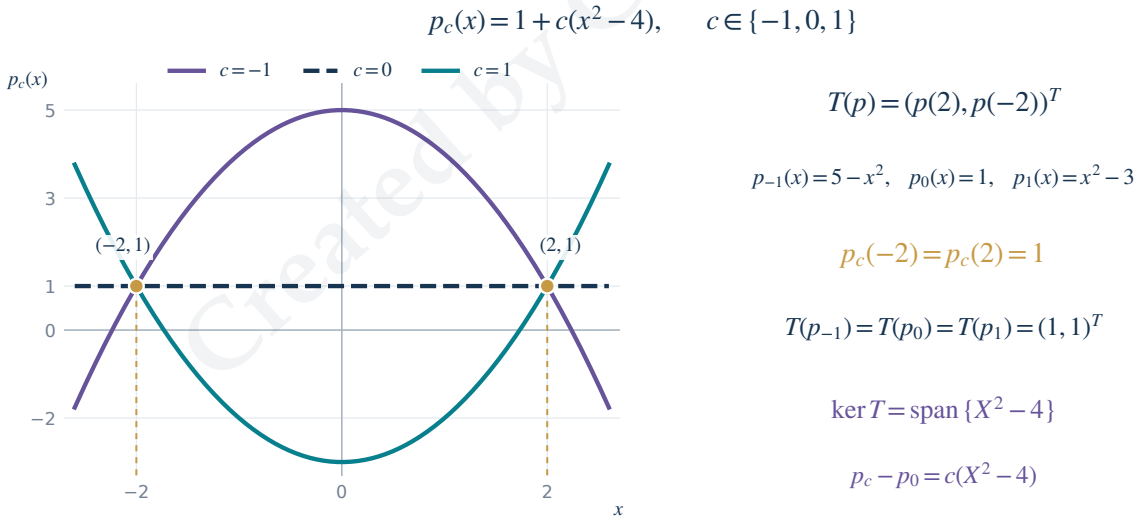


Figure 11.2. Three distinct polynomials give the same two measurements: $p_c(2) = p_c(-2) = 1$ for $c = -1, 0, 1$. Their differences $p_c - p_0 = c(X^2 - 4)$ lie in the kernel of $T(p) = (p(2), p(-2))^T$, so these measurements do not distinguish the polynomials.

Created by Chen Gadi

 © 2026 Chen Gadi

EXERCISE 11.20 CHALLENGE

Let V, W, U be finite-dimensional spaces over a field F, and let $S : V \to W$ and $T : V \to U$ be linear. Please prove that $\ker S \subseteq \ker T$ if and only if there is a linear map $R : W \to U$ with $T = R \circ S$. When S is onto, please prove that R is unique.

WORKED SOLUTION

If $T = R \circ S$ and $x \in \ker S$, then $T(x) = R(0) = 0$, so $x \in \ker T$. Conversely, assume the kernel inclusion and define on $\operatorname{Im} S$

$$R_0(Sx) = Tx.$$

This is well-defined: if $Sx = Sy$, then $x - y \in \ker S \subseteq \ker T$, so $Tx = Ty$. Also

$$R_0(aSx + bSy) = R_0(S(ax + by)) = T(ax + by) = aTx + bTy,$$

which proves linearity. Choose a basis of $\operatorname{Im} S$ and extend it to a basis of W. Define R by R_0 on the first basis vectors and by zero on the added vectors, then extend linearly. This gives $R(Sx) = Tx$ for every x. If S is onto, every $w \in W$ is Sx for some x. The equality forces $R(w) = Tx$, so there is only one possible map R.

EXERCISE 11.21 CHALLENGE

$$A \in \mathbb{F}^{m\times n}, \quad B \in \mathbb{F}^{n\times p}.$$

$$\operatorname{rank}(AB) = \operatorname{rank} B - \dim(\operatorname{Col} B \cap \ker A).$$

$$A_0 = \begin{pmatrix} 1 & 0 & -2 \\ 0 & 1 & 1 \end{pmatrix}, \qquad B_0 = \begin{pmatrix} 2 & 0 \\ 0 & 1 \\ 1 & 0 \end{pmatrix} \quad \text{over } \mathbb{R}.$$

Please prove the identity and please deduce $\operatorname{rank}(AB) \geq \operatorname{rank} A + \operatorname{rank} B - n$. Please verify the identity for A_0, B_0.

WORKED SOLUTION

Apply rank-nullity to the restriction L. Its image and kernel are exactly the spaces in the identity. In the example, the kernel of A_0 lies in the column space of B_0, so their intersection has dimension 1.

$$L : \operatorname{Col} B \to \mathbb{F}^m, \qquad L(y) = Ay, \quad \operatorname{Im} L = \operatorname{Col}(AB), \quad \ker L = \operatorname{Col} B \cap \ker A.$$

$$\operatorname{rank} B = \dim \operatorname{Im} L + \dim \ker L, \qquad \dim \ker L \leq \dim \ker A = n - \operatorname{rank} A.$$

$$\ker A_0 = \operatorname{span}\{(2, -1, 1)^T\}, \quad (2, -1, 1)^T = B_0 e_1 - B_0 e_2, \quad \operatorname{rank} B_0 = 2,$$

$$A_0 B_0 = \begin{pmatrix} 0 & 0 \\ 1 & 1 \end{pmatrix}, \qquad \operatorname{rank}(A_0 B_0) = 1 = 2 - 1.$$

Created by Chen Gadi

 © 2026 Chen Gadi